Tutorial 2

Ex 8.2

Tabular integration

	diff	int
$+$	x^2-2x+1	e^{2x}
$\boxed{-}$	$2x-2$	$\frac{1}{2}e^{2x}$
$+$	2	$\frac{1}{4}e^{2x}$
$-$	0	$\frac{1}{8}e^{2x}$

Q10 $\displaystyle\int \underbrace{(x^2-2x+1)}_{\text{diff}}\underbrace{e^{2x}}_{\text{int}}\,dx$

$$= (x^2-2x+1)\left(\tfrac{1}{2}e^{2x}\right) - (2x-2)\left(\tfrac{1}{4}e^{2x}\right) + 2\cdot\tfrac{1}{8}e^{2x} + C$$

$$= \cdots + C$$

Q24 $\boxed{\int \underbrace{e^{-2x}}_{\text{int}} \underbrace{\sin 2x}_{\text{diff}} \, dx}$

	diff	int
$+$	$\sin 2x$	e^{-2x}
$-$	$2\cos 2x$	$-\frac{1}{2}e^{-2x}$
$+$	$-4\sin 2x$	$\frac{1}{4}e^{-2x}$

$$\int (-4\sin 2x)\left(\tfrac{1}{4}e^{-2x}\right) dx$$

$$= -\frac{1}{2}e^{-2x}\sin 2x$$

$$-\frac{1}{2}e^{-2x}\cos 2x$$

$$-\int \sin 2x \cdot e^{-2x} \, dx$$

Let $I := \int e^{-2x} \sin 2x \, dx$

$$\Rightarrow I = -\frac{1}{2}e^{-2x}\sin 2x - \frac{1}{2}e^{-2x}\cos 2x - I$$

$$I = \frac{1}{2}\left(\quad \cdots \quad \right) + C$$

Q42 $\int \underbrace{\sin 2x}_{\text{diff}} \underbrace{\cos 4x}_{\text{int}} \, dx$

① Product to Sum

$$\sin A \cos B = \frac{1}{2}\sin(A-B) + \frac{1}{2}\sin(A+B)$$

② Int. by part.

	diff	int
+	$\sin 2x$	$\cos 4x$
−	$2\cos 2x$	$\frac{1}{4}\sin 4x$
+	$-4\sin 2x$	$-\frac{1}{16}\cos 4x$

$$\int \sin 2x \cos 4x \, dx$$

$$= \frac{1}{4}\sin 2x \sin 4x + \frac{1}{8}\cos 2x \cos 4x + \int \frac{1}{4} \sin 2x \cos 4x \, dx$$

$$I := \int \sin 2x \cos 4x \, dx$$

$$\Rightarrow I = \cdots + \frac{1}{4} I$$

$$\Rightarrow I = \cdots + C$$

Q52 $\displaystyle\int \underset{\text{int}}{x^2}\,\underset{\text{diff}}{\tan^{-1}\frac{x}{2}}\,dx$

	diff	int
$+$	$\tan^{-1}\frac{x}{2}$	x^2
$-$	$\frac{1}{2}\cdot\frac{1}{1+\left(\frac{x}{2}\right)^2} = \frac{1}{2+\frac{x^2}{2}}$	$\frac{x^3}{3}$

$$= \frac{x^3}{3}\tan^{-1}\frac{x}{2} - \int \frac{1}{2+\frac{x^2}{2}}\cdot\frac{x^3}{3}\,dx$$

$$\frac{1}{2+\frac{x^2}{2}}\cdot\frac{x^3}{3} = \frac{x^3}{\left(2+\frac{x^2}{2}\right)3} = \frac{2x^3}{(4+x^2)3} = \frac{2}{3}\cdot\boxed{\frac{x^3}{4+x^2}}$$

$\frac{2}{2}$

$$x(4+x^2) = 4x + x^3$$

$$x(x^2+4)$$

$$\frac{x^3}{4+x^2} = \frac{x^3+4x-4x}{4+x^2} = \frac{x^3+4x}{\underbrace{4+x^2}_{x}} - \frac{4\boxed{x}}{\boxed{4+x^2}}$$

$u = 4+x^2$

$du = 2\boxed{x\,dx}$

Q53 $\int (1+2x^2)e^{x^2}\,dx$

$u = x^2$

$du = \boxed{2x}\,dx$

$\int (1+2x^2)e^{x^2}\,dx$

$= \int \boxed{(1+2x^2)\frac{1}{2x}}\,e^{u}\,du$

$= \int 1 \cdot e^{x^2}\,dx + \int 2x^2 e^{x^2}\,dx$

	diff	int
+	e^{x^2}	1
−	$2xe^{x^2}$	x

~~Integrat~~ $\int e^{x^2}\,dx = xe^{x^2} - \int 2x^2 e^{x^2}\,dx$

$\therefore \int (1+2x^2)e^{x^2}\,dx = xe^{x^2} + C$

Try Q54

Ex. 8.3

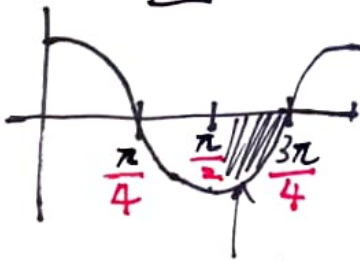
Q17 $\int_0^{\pi} 8\sin^4 x\,dx = \int_0^{\pi} 8(\sin^2 x)^2\,dx$

$(\sin x + \cos x)^2$ $\boxed{\sin^2 x + \cos^2 x \quad 1 - 2\sin x\cos x}$

$\sin^2 x = \dfrac{1-\cos 2x}{2}$

Q30 $\int_{\pi/2}^{3\pi/4} \sqrt{1-\sin 2x}\,dx \cdot \sqrt{\dfrac{1+\sin 2x}{1+\sin 2x}}$

$\cos \boxed{2x}$



negative

$= \int_{\pi/2}^{3\pi/4} \dfrac{\sqrt{1-\sin^2 2x}}{\sqrt{1+\sin 2x}}$

$1-\sin^2\theta = \cos^2\theta$

$\sqrt{1-\sin^2 2x} = \sqrt{\cos^2 2x}$

$= |\cos 2x|$

$= \int_{\pi/2}^{3\pi/4} \dfrac{|\cos 2x|}{\sqrt{1+\sin 2x}}\,dx$

$|\cos 2x| = \begin{cases} \cos 2x & \text{if } \cos 2x \geq 0 \\ -\cos 2x & \text{if } \cos 2x \leq 0 \end{cases}$

$= \int_{\pi/2}^{3\pi/4} \dfrac{-\cos 2x}{\sqrt{1+\sin 2x}}\,dx$

$u = 1+\sin 2x \Rightarrow du = 2\cos 2x\,dx$

Q31 $\int_0^{\pi/2} \theta\sqrt{1-\cos 2\theta}\, d\theta$

① Mult. by conjugate

② Double angle: $1-\cos 2\theta = \underline{2\sin^2\theta}$

$\boxed{1} \pm \sin 2\theta = \sin^2\theta + \cos^2\theta \pm 2\sin\theta\cos\theta$

$= (\sin\theta \pm \cos\theta)^2$

Avoid improper integral

Ex 8.4

$\sqrt{x^2-1} \rightarrow x = \sec\theta \Rightarrow \underline{0 \leq \theta < \frac{\pi}{2}}$

$\sqrt{1+x^2} \rightarrow x = \tan\theta \Rightarrow \sim$

$\sqrt{1-x^2} \rightarrow x = \sin\theta \Rightarrow \sim$

Q48 $\displaystyle\int \frac{\sqrt{x-2}}{\sqrt{x-1}}\,dx$ $\qquad u = x-1 \Rightarrow du = dx$

$$= \int \frac{\sqrt{u-1}}{\sqrt{u}}\,du \qquad t = \sqrt{u} \Rightarrow dt = \frac{1}{2\sqrt{u}}\,du$$

$$\Downarrow$$

$$u = t^2$$

$$= 2\int \sqrt{u-1}\,dt$$

$$= 2\int \sqrt{t^2-1}\,dt \qquad t = \sec\theta,\ 0 \leq \theta < \frac{\pi}{2}$$

Q45

$$\int \sqrt{\frac{4-x}{x}}\,dx \qquad \frac{\sqrt{4-x}}{\sqrt{x}} \qquad u=\sqrt{x}$$